



University of St.Gallen

Institute of Computer Science



ETRA 2026

EYE-CARE:

Eye Tracking for Context-Adaptive Systems Based on Activity and Cognitive Load Prediction

June 1st - 4th 2026, Marrakech, Morocco

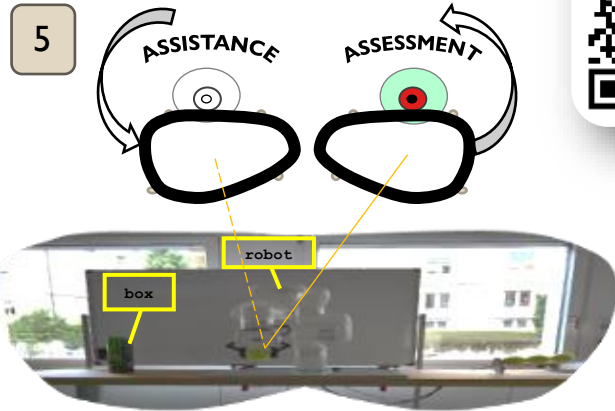
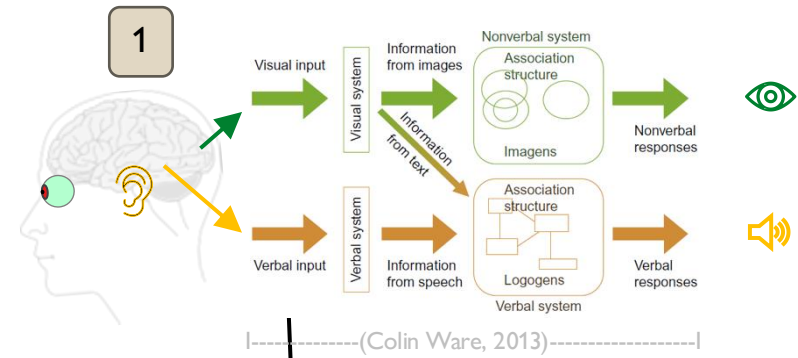
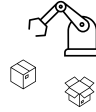
Kenan Bektaş and Amine Abbad-Andaloussi

From insight to impact.



Kenan studies . . .

1. Dual Coding Theory ([Allan Paivio](#))
2. Cognitive Load (CL) Theory ([John Sweller](#))
3. Cognitive Offloading ([Risko and Gilbert](#))
4. Human Augmentation ([Raisamo et al.2019](#); [Villa et al., 2023](#))
5. Pervasive Eye Tracking ([Bulling and Gellersen, 2010](#); [Bektas, 2020](#))



Motor-
Sensory-
Cognitive-
Augmentation



4

“Technologies
that help users overcome limitations
and integrate with their body”

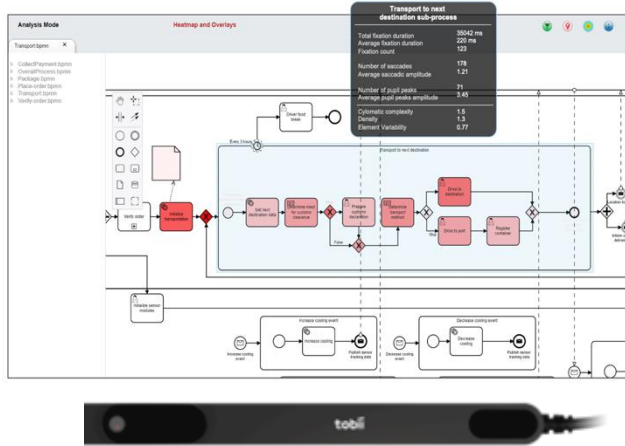
- 2
- Intrinsic CL
 - Extraneous CL
 - Germane CL

3

“Use **physical action**
to alter the **information**
processing
requirements of a task
to **reduce**
cognitive demand”

Toward A Pervasive Gaze-Contingent Assistance System:
Attention and Context-Awareness in Augmented Reality

Eye-tracking Methods & Frameworks



Context-Adaptive Systems



1.) Biological Signal (e.g., EEG, pupil dilation, HRV, skin conductance) is recorded from the user

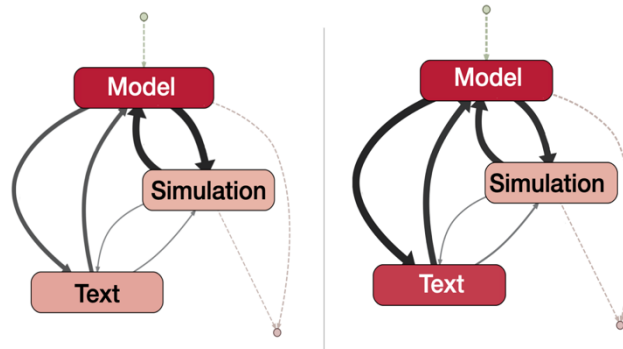


2.) Biological signal is analyzed to derive a cognitive/behavioral state



3.) System adapts on the base of the user's cognitive/behavioral state

Cognitive & Behavioral Aspects



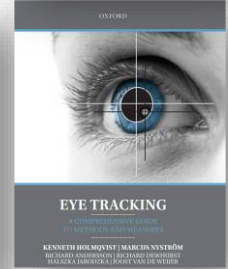
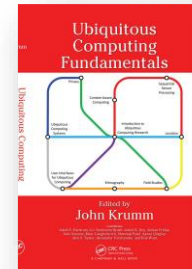
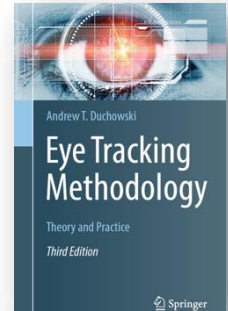
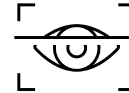
Learn more about my research in this stream

Ubiquitous Computing and Context-Aware Computing

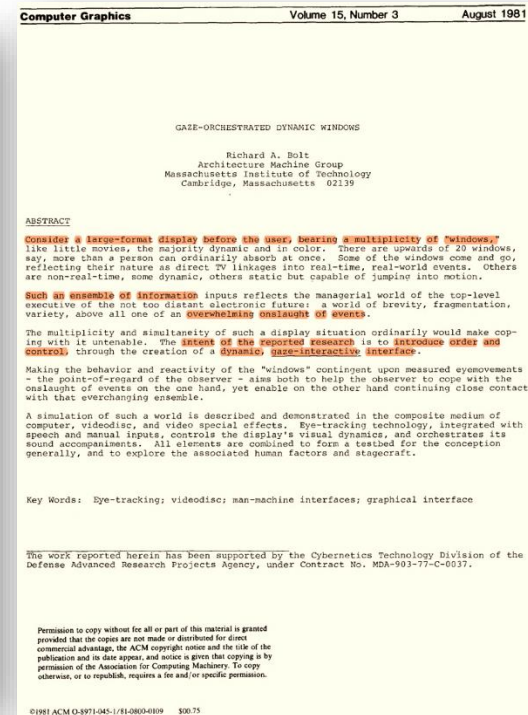


Seamless integration of networked (micro-) computers and displays into the physical world.

Systems that can adapt to the **User**, its **Physical-** and **Computational-**environment

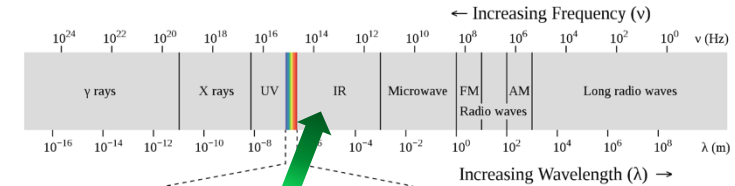


Put It There



Richard Bolt and colleagues in 80's (MIT)

Eye Trackers (and MR-headsets)



Mixed Reality (MR) Headsets

Eye Tracking (ET) Devices

- Mobility
- Weight or Size
- Field-of-view
- Rendering quality
- Display resolution
 - Frame rate
- Connectivity
- Battery life
- App development
 -

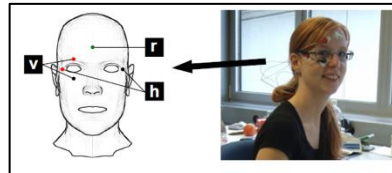
Connectivity	Wired	Mobile or Wireless		Wired
Examples	HTC VIVE Pro Eye, Varjo XR3	Microsoft HoloLens 2	Tobii Glasses 3, Pupil Core	Tobii Pro Fusion, EyeLink 1000 Plus
				
ET Sampling Rate	120-200Hz	30Hz	50-200Hz	60-2000Hz
ET Accuracy	≈ 1.1°	≈ 1.5° - 3.0°	≈ 0.6° - 0.8°	≈ 0.2° - 0.5°

Head-Mounted

Remote
or
Desktop-Mounted

- Mobility
- Weight or Size
- Field-of-view
- Data quality or accuracy
 - Sampling rate
- Display resolution
 - Connectivity
- Battery life
- Open-source?
 -

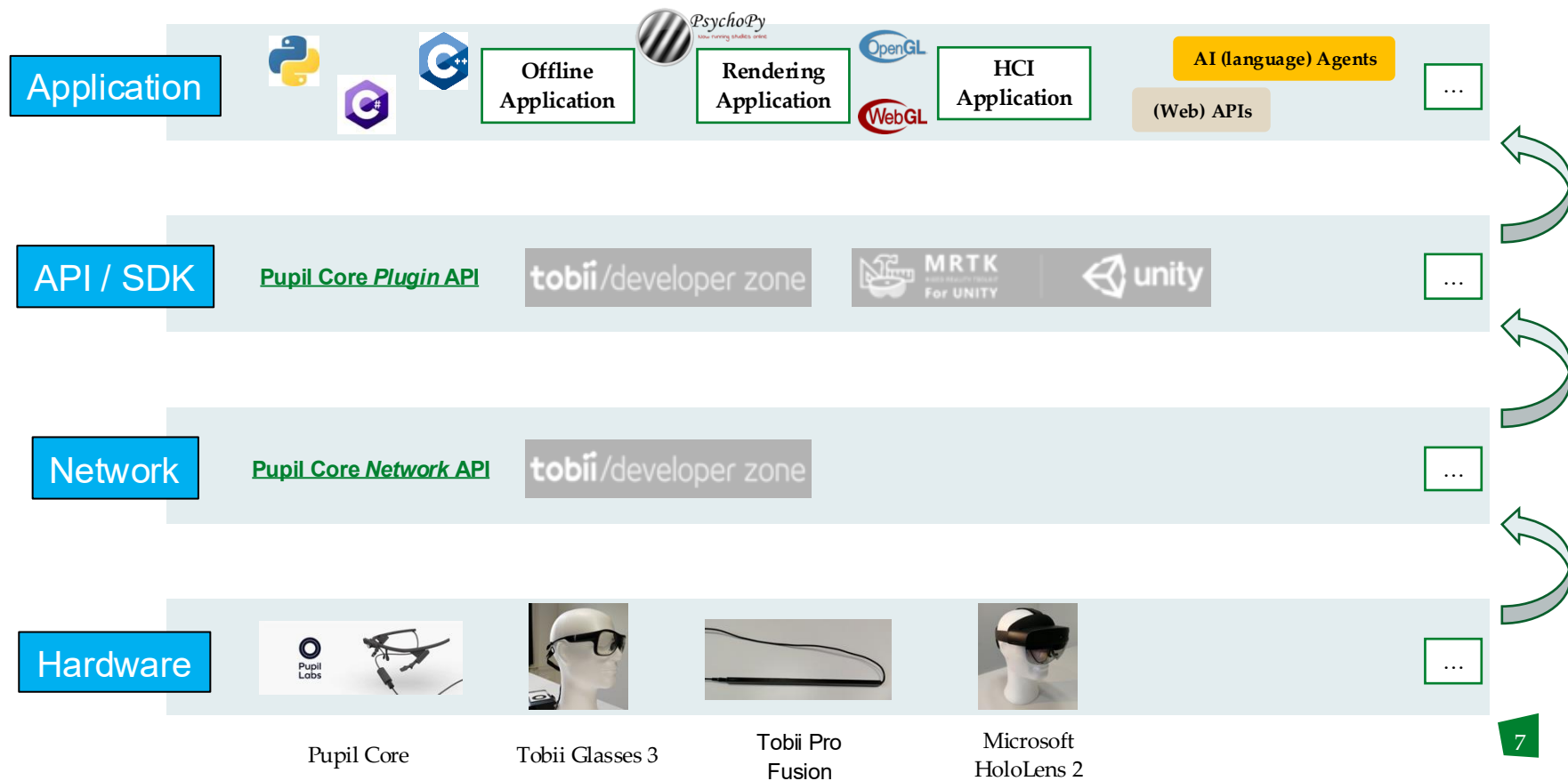
Bektaş, K., Strecker, J., Mayer, S., & Stolze, M. (2022). [EToS-1: Eye Tracking on Shopfloors for User Engagement with Automation](#). In *AutomationXP22: Engaging with Automation, CHI'22*



Eye Movement Analysis for Activity Recognition
Using Electrooculography

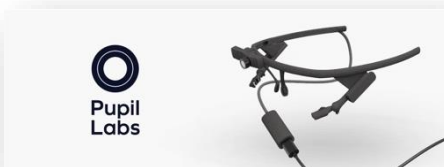
Andreas Bulling, Student Member, IEEE, Jamie A. Ward,
Hans Gellersen, and Gerhard Tröster, Senior Member, IEEE

Hardware and Software



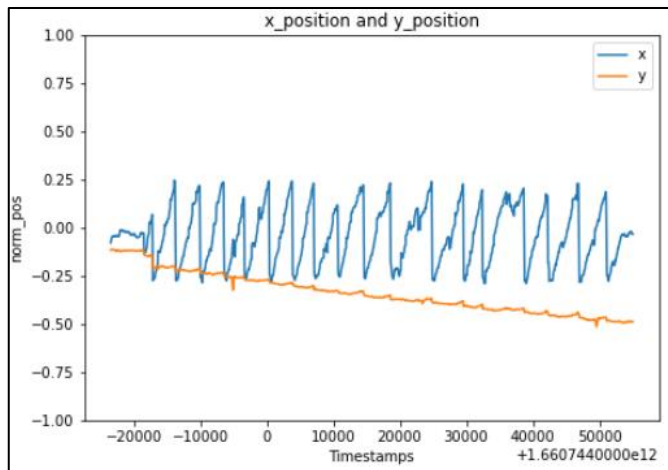
What is Eye Tracking Data?

Eye tracker



Time series data

	eyeDataTimestamp	gazeHasValue	gazeOrigin_x	gazeOrigin_y	gazeOrigin_x
1	1660743976560	True	0.00688	0.06962	0.00688
2	1660743976594	True	0.00616	0.06952	0.00616
3	1660743976627	True	0.00529	0.06944	0.00529
4	1660743976660	True	0.00506	0.06944	0.00506
5	1660743976694	True	0.00494	0.06947	0.00494
6	1660743976727	True	0.00470	0.06947	0.00470
7	1660743976760	True	0.00399	0.06940	0.00399
8	1660743976794	True	0.00281	0.06946	0.00281
9	1660743976827	True	0.00191	0.06958	0.00191
10	1660743976860	True	0.00132	0.06972	0.00132



Activity Prediction from Eye Movements

1. Data Collection

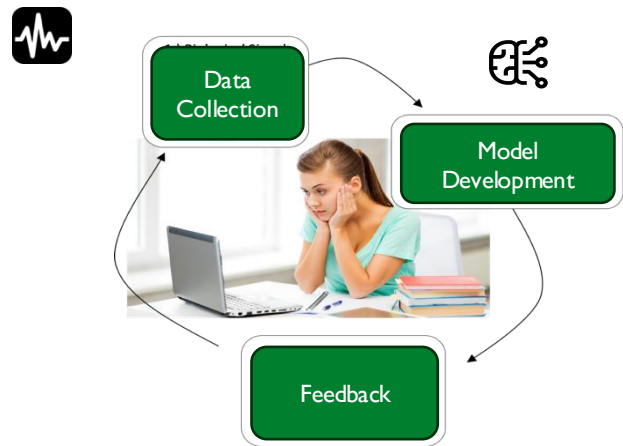
- Understand the **purpose**: Diagnostics, Rendering, HCI. For example, *Reading* material (paper or digital etc.), geometric constraints (desktop vs. mobile), static vs. dynamic scene (single page vs. scrolling) etc.
- Understand the **temporal** constraints that are relevant for the purpose (seconds to hours)
- Collect ground truth data

2. Model Development

- Calculate eye movement metrics (fixation, saccade, blinks, etc.)
- Data Preprocessing (cleaning, smoothing, interpolation, etc.)
- Calculate features (normalization)
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3. Feedback

- What do we do with the detected activity?



Notebook 1:

📁 Data/RawGazeData	Delete Data/RawGazeData/placeholder.txt
📁 demo_images	intervention example added
📄 AnSVMClassifierForHL2GazeFeatures.ipynb	Add files via upload
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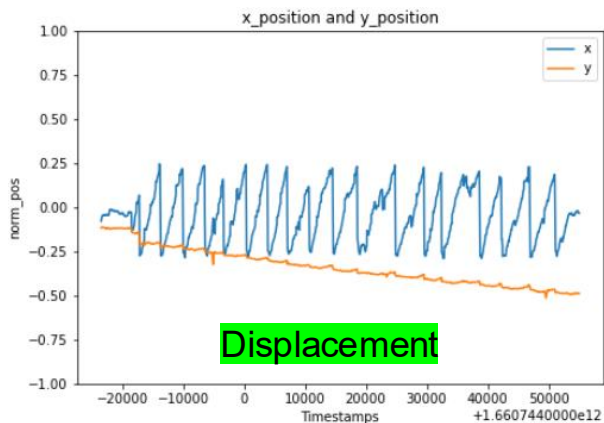


Eye Movement Measures



Time Series

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Salvucci and Goldberg 2000

I-DT (protocol, dispersion threshold, duration threshold)

While there are still points

Initialize window over first points to cover the duration threshold

If dispersion of window points $\leq$ threshold

Add additional points to the window until dispersion $>$ threshold

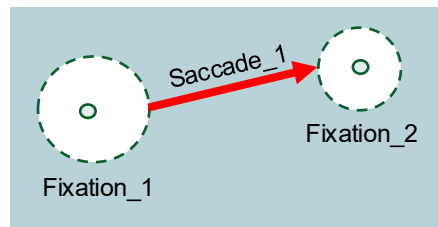
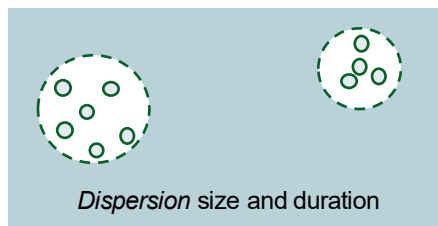
Note a fixation at the centroid of the window points

Remove window points from points

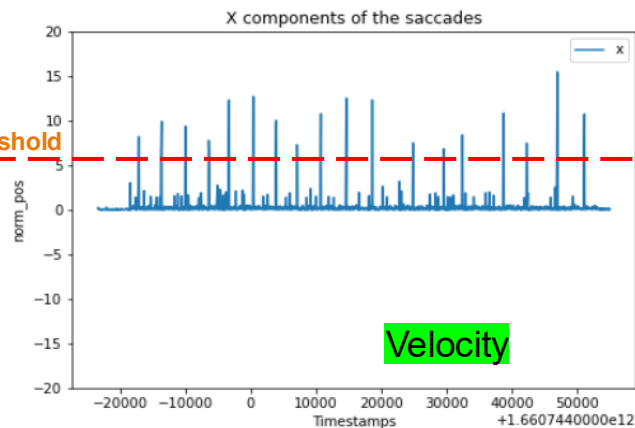
Else

Remove first point from points

Return fixations

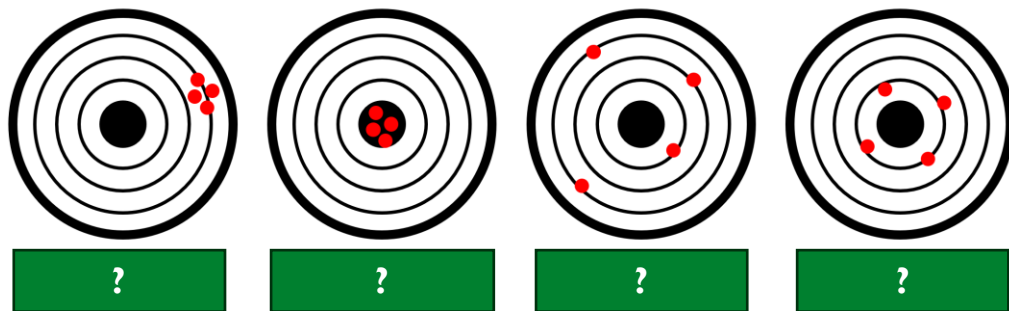


Velocity Threshold



Eye Tracking Data

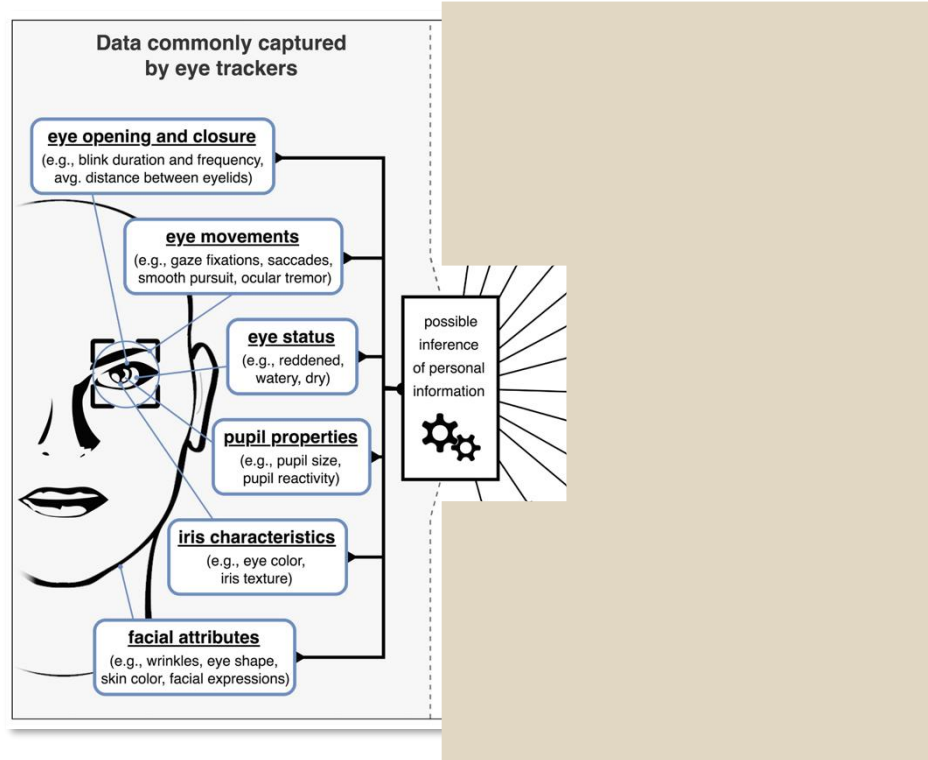
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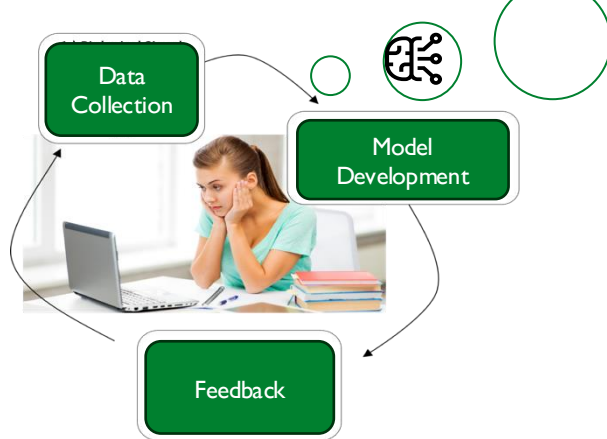
Kasneji et al 2024 <https://arxiv.org/abs/2404.15435>

- **Sampling rate**
 - *How many raw data per second?*
 - Measured in Hertz (Hz)
- **Accuracy**
 - *Deviation between the actual and measured*
 - Measured in degrees °
 - Depends on calibration quality and hardware
- **Precision**
 - *Deviation of the measurements*
 - Measured by the root mean square (RMS) error
 - Mainly hardware dependent
- **Calibration**
 - *A mapping between the measured and actual points of interest*
 - Pay attention to *slippage* (compensation) and *lighting* conditions

What can **You** infer from Eye Tracking Data?

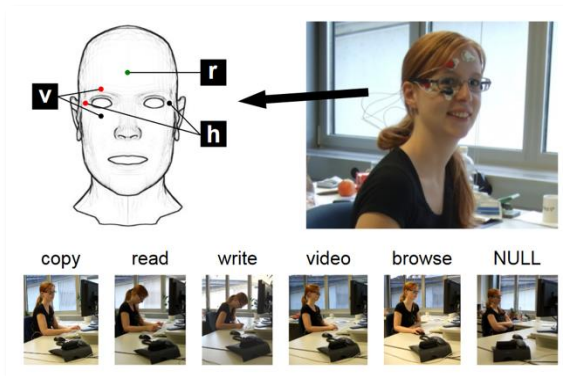
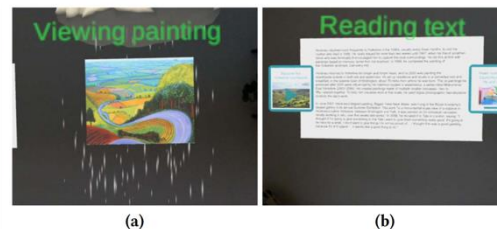


Why do **You** use Eye Tracking Data?



Keep an **Eye** on your **Activities**

Eye Tracking can also be used to infer someone's activity!

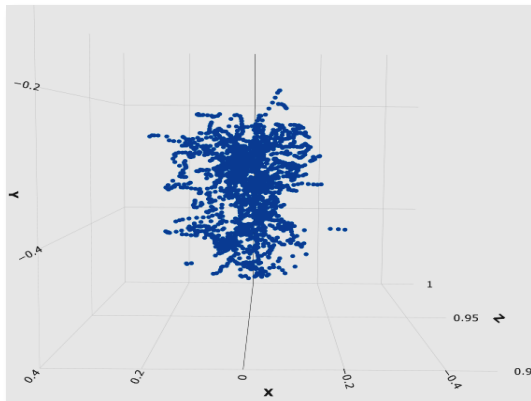
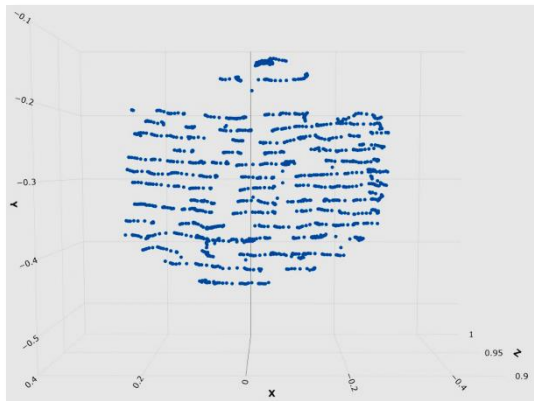


- [1] Bulling et al. 2011. Eye Movement Analysis for Activity Recognition Using Electrooculography. IEEE TPAMI 33, 4 (April 2011), 741–753. <https://doi.org/10.1109/TPAMI.2010.86>
- [2] Scargill et al. 2022. Demo Abstract: Catch My Eye: Gaze-Based Activity Recognition in an Augmented Reality Art Gallery. IPSN'22. <https://doi.org/10.1109/IPSIN54338.2022.00052>
- [3] David-John et al. 2021. Towards gaze-based prediction of the intent to interact in virtual reality. ETRA'21. <https://doi.org/10.1145/3448018.3458008>

Offline Gaze Data Analysis for Activity Recognition



Which gaze data belongs to which activity?



Keep an Eye on our Activities 😊

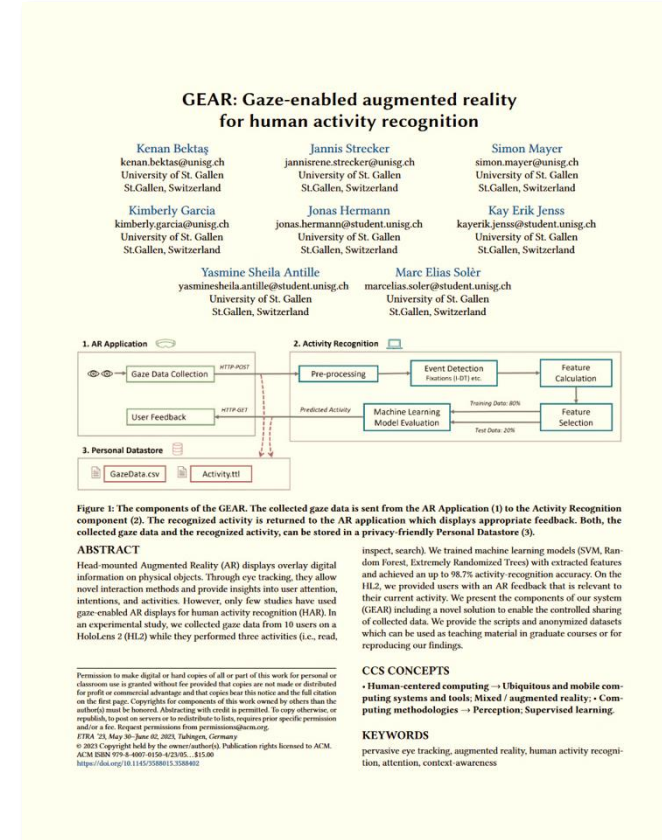
Output from three years ago:

1st Activity: Reading

Link to video: <https://www.youtube.com/watch?v=Dq-Z5p61J8E>

Kenan Bektaş, Jannis Strecker, Simon Mayer, Kimberly Garcia, Jonas Hermann, Kay Erik Jenß, Yasmine Sheila Antille, and Marc Solèr. 2023. GEAR: Gaze-enabled augmented reality for human activity recognition. In Proceedings of the 2023 Symposium on Eye Tracking Research and Applications (ETRA '23).

<https://doi.org/10.1145/3588015.3588402>

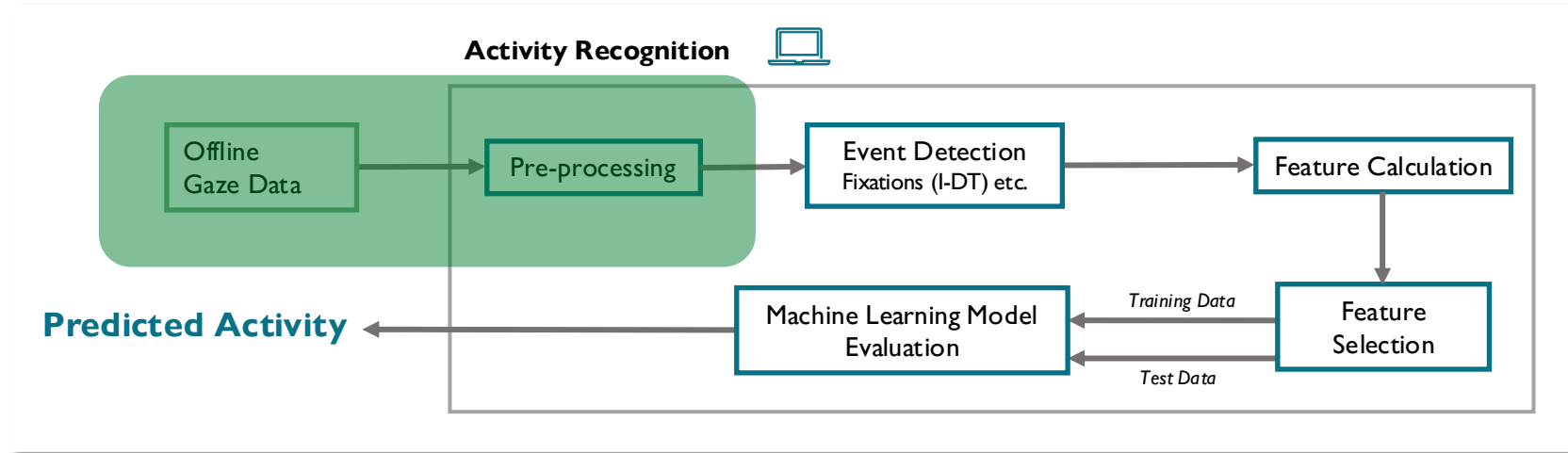


Offline Gaze Data Analysis for Activity Recognition



Develop a model that classifies given eye movement data to estimate the current activity of a user.

Pipeline:



From Eye Movement Metrics to Features

Type	Duration (ms)	Amplitude	Velocity	Example
Fixation	200-300	-	-	When the eye temporarily stops at a word during reading.
Saccade	30-80	4-20°	30-500°	When the eye moves from one object to another.
(Blink)	-	-	-	When the eye is closed for a short moment of time.

Eye movement features



Descriptive statistics of the eye movement metrics e.g.,

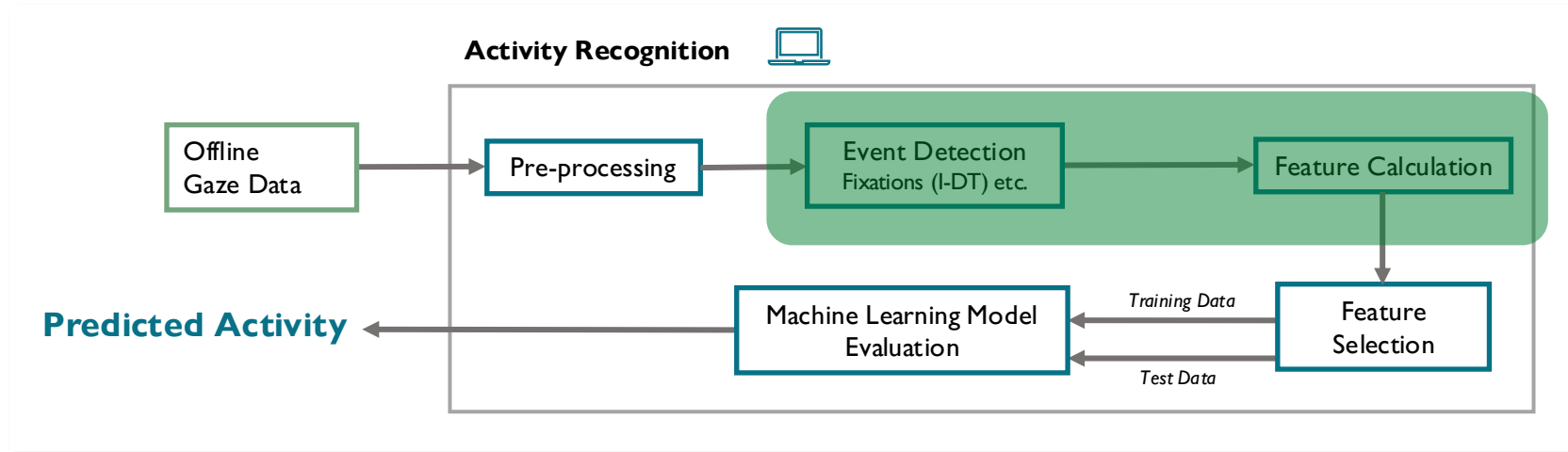
- (min/max/avg) fixation duration
- fixation dispersion
- fixation density
- blink rate per minute

Offline Gaze Data Analysis for Activity Recognition



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Pipeline:



Activity Prediction from Eye Movements

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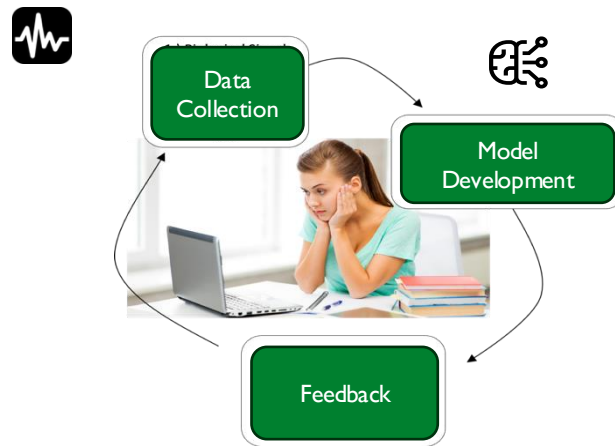
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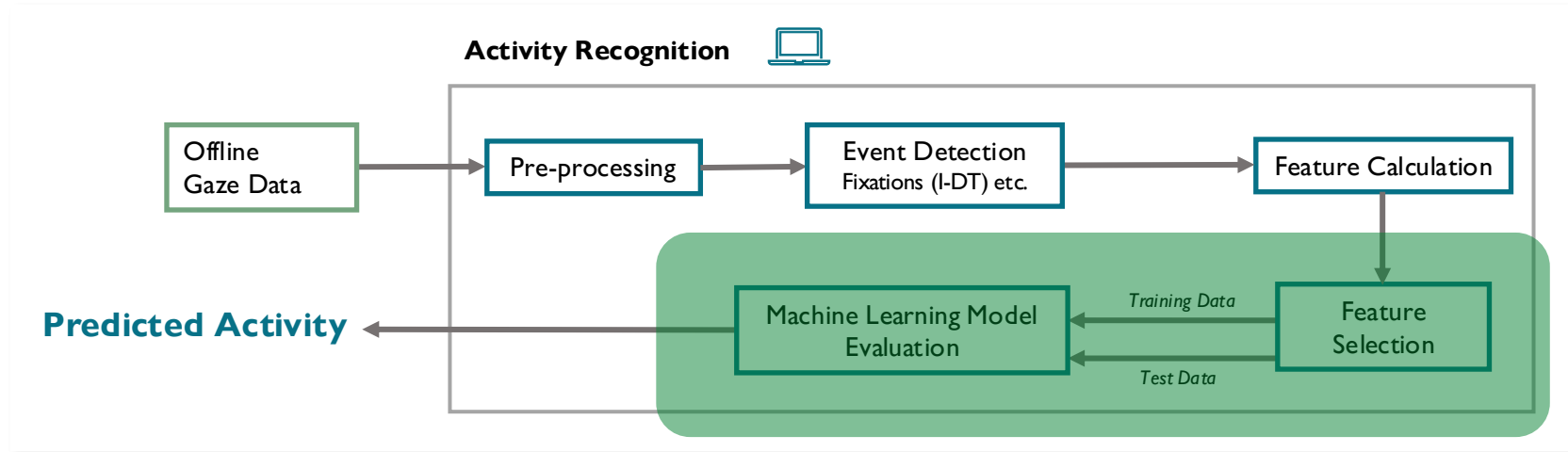
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Offline Gaze Data Analysis for Activity Recognition



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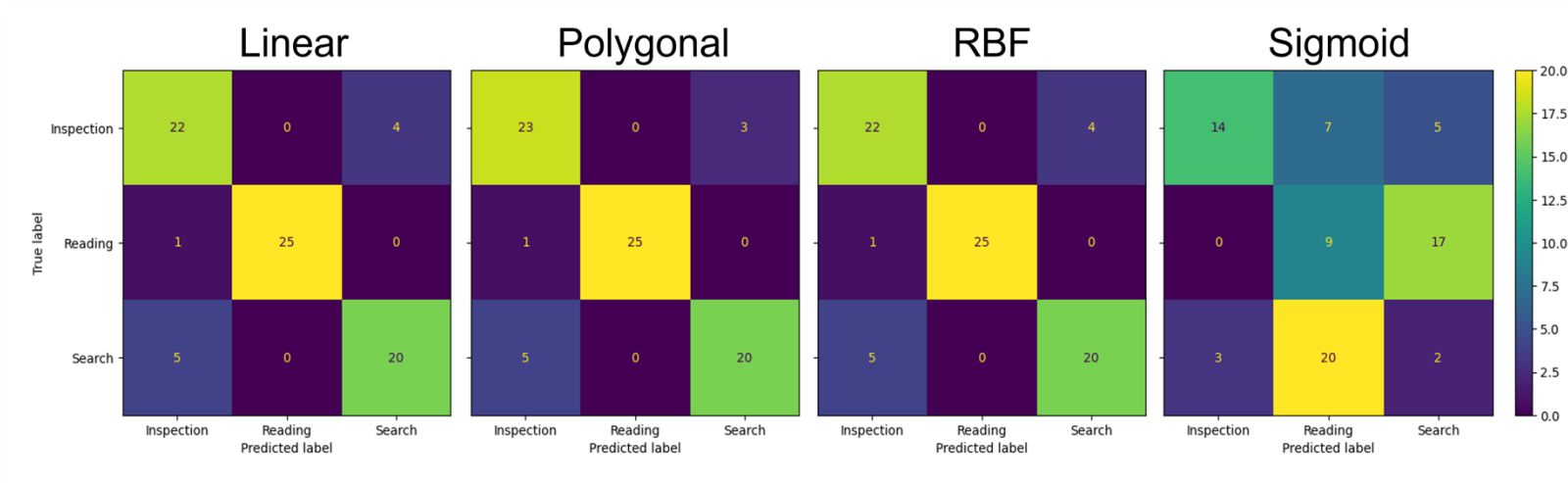
Notebook 3:



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The existing SVM-based classifier

Confusion Matrices:



Accuracies:

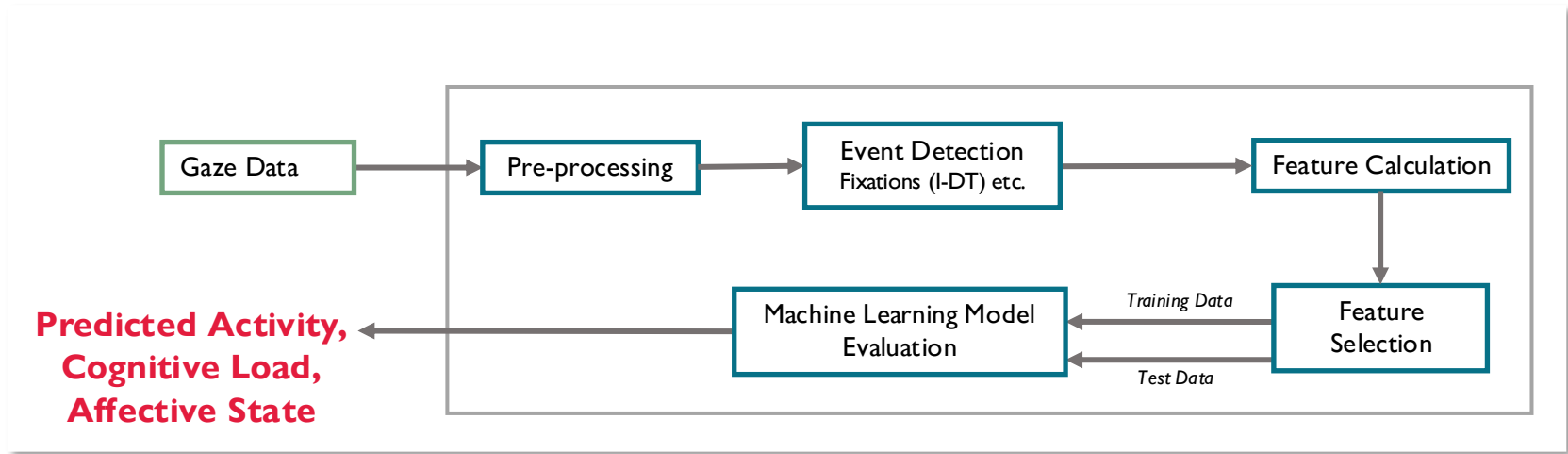
87.0%

88.3%

87.0%

32.4%

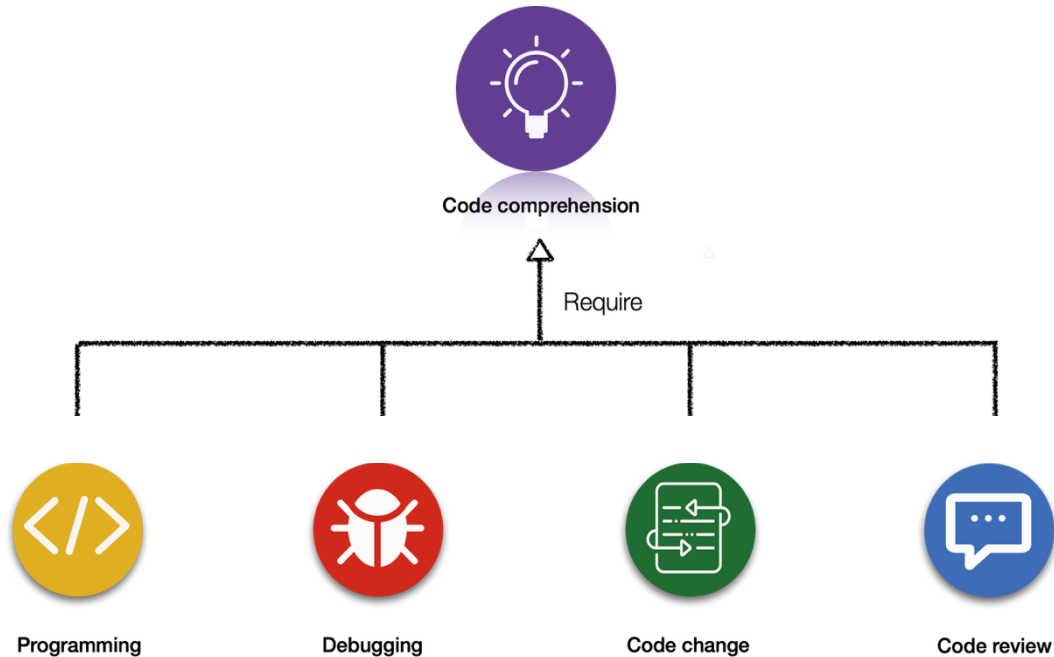
Generalizing Beyond Activity Recognition



Similar pipeline. Different data; different output

Another case from the literature: Cognitive Load Detection During Program Comprehension

The Case of Cognitive Load Detection During Program Comprehension: Context



The Case of Cognitive Load Detection During Program Comprehension: Context



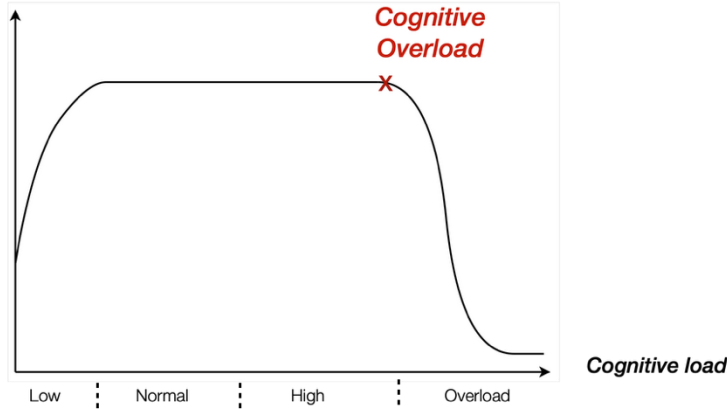
Code comprehension

Can be estimated using



Cognitive load measures

Task performance
e.g., Code comprehension



Problem: cognitive overload **hinders** code comprehension

Adapted from [1]

Cognitive Load Measures



Eye-tracking – Light weight, continuous measure

```
8 public static void main(String[] args) {
9     List<Object> array = new ArrayList<Object>();
10
11     Object r = new Rectangle();
12     array.add(r);
13     Object e = new Triangle();
14     Object s = new Circle();
15     array.add(s);
16     array.add(e);
17
18     for(int i=0; i<array.size(); i = next(i,array)){
19         Graphics.draw(array.get(i));
20     }
21 }
22
23 public static int next(int i, List<Object> array) {
24     if(array.get(i) instanceof Triangle) return array.size();
25     else if(array.get(i) instanceof Rectangle) return i+2;
26     return i-1;
27 }
28
```

Pupil measures

Fixation measures

Saccade measures

Scan-path measures

A Coarse-Grained Approach

```
8= public static void main(String[] args) {
9     List<Object> array = new ArrayList<Object>();
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```



Estimate cognitive load at the level of the **entire source-code**

e.g., (Fritz et al. 2014)

Challenges:

- Not all the parts of the code have the same level of difficulty
- One **cannot pinpoint** the mentally demanding parts of code

A Fine-Grained Approach

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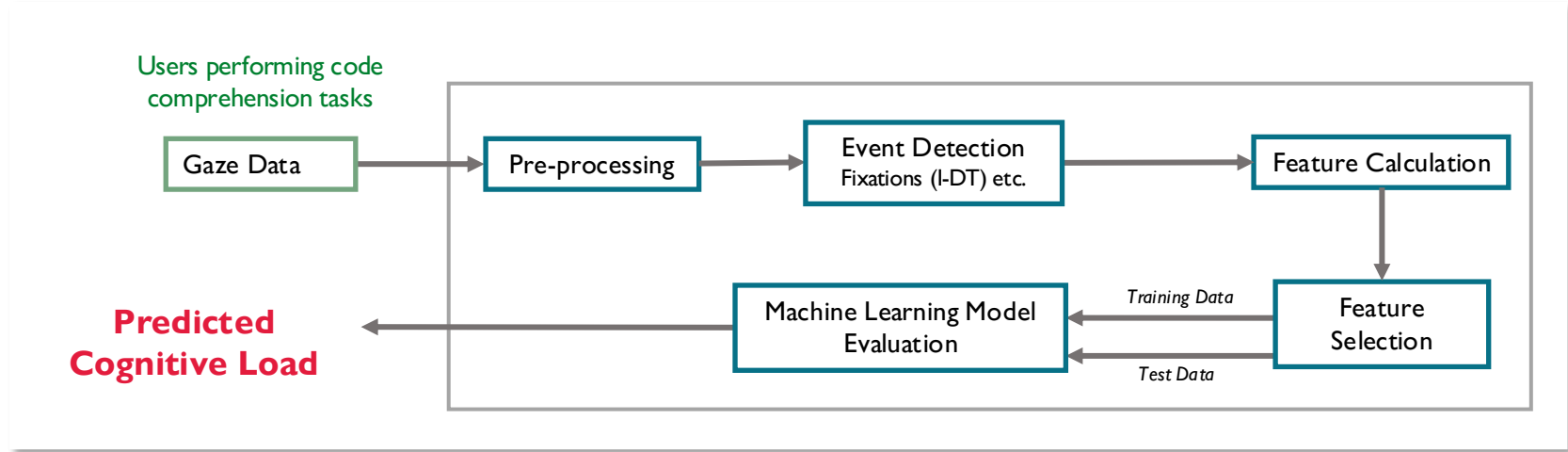
Estimate cognitive load at a fine-grained level e.g., each line of code

e.g., (Abbad-Andaloussi et al. 2022)

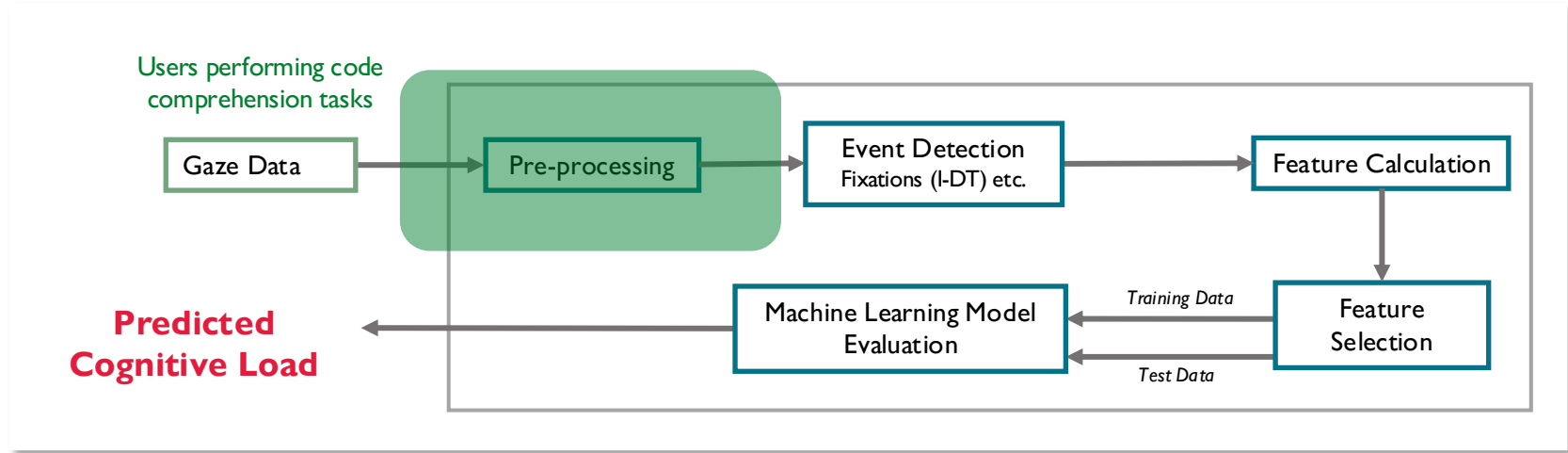
Benefits:

- Pinpoint the exact parts that are mentally demanding in the source-code
- Subsequently, focus on the mentally demanding parts of code during refactoring and code review tasks to enhance the comprehensibility of the source-code

Back to Our Pipeline



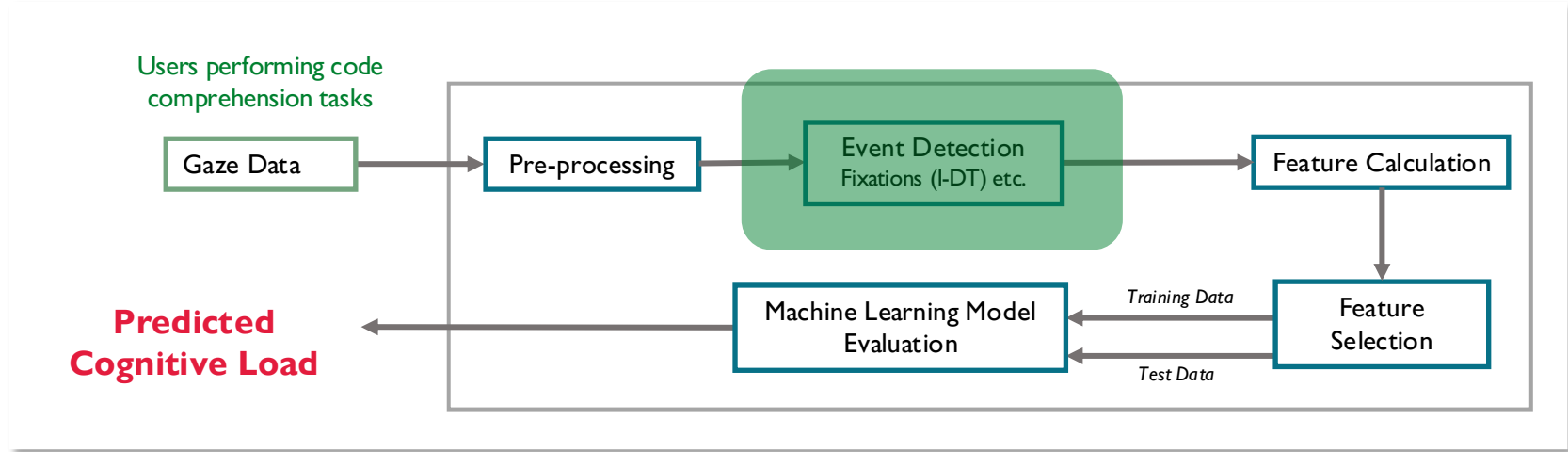
Back to Our Pipeline



Pre-processing

- **Blink detection:** Detect and remove blink artifacts.
- **Standard deviation:** Remove outliers identified by statistics.
- **Linear interpolation:** Linearly interpolate missing values for gaze and pupil data.
- **Butterworth Filter:** Applies a third order lowpass Butterworth filter.

Back to Our Pipeline



Event Detection

Using the Velocity-Threshold Identification (I-VT) fixation classification algorithm

Interpolation: ☒

Max gap length in ms:

75

Noise reduction: ☐

Method:

Moving median



Window size in samples:

3

Fixation filter:

Window length in ms:

20

Velocity threshold in
degrees/second:

30

Discard short fixations: ☒

Minimum fixation duration in
ms:

60

Merge adjacent fixations: ☒

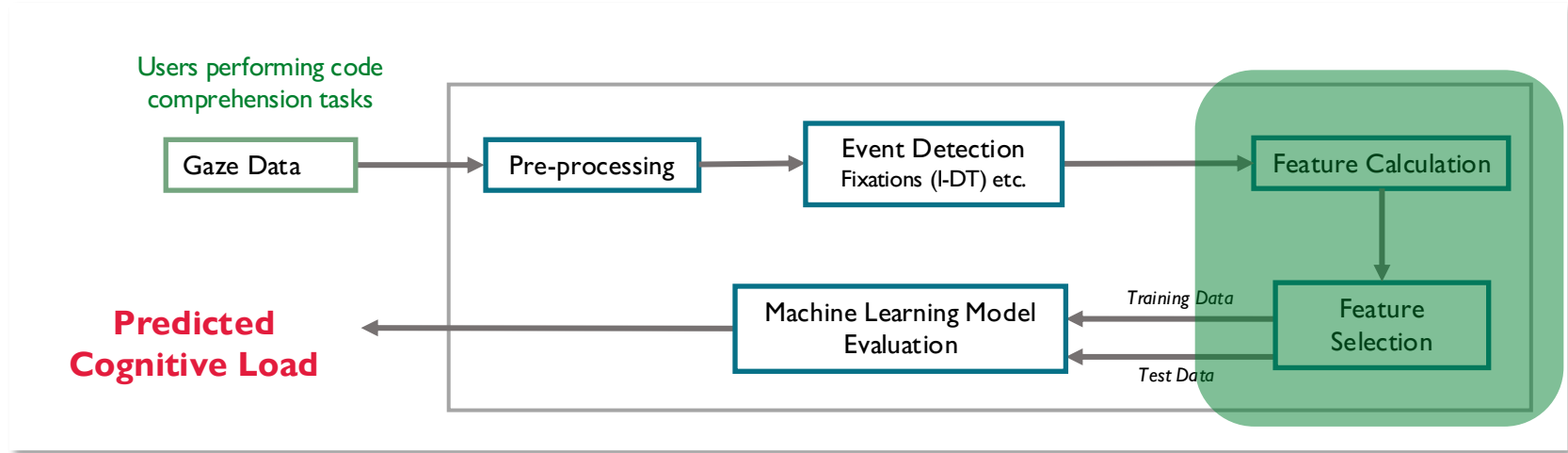
Maximum time between
fixations in ms:

75

Maximum angle between
fixations in degree:

0.5

Back to Our Pipeline



Feature Calculation and Selection

Category	Feature	Description
Fixation	Fixation duration	Duration of the fixation in milliseconds
	Fixation duration <i>fix.</i> $\geq 250ms$	<i>Expression:</i> fixation duration if fixation duration $\geq 250ms$ else NaN
	Fixation duration <i>fix.</i> $\geq 500ms$	<i>Expression:</i> fixation duration if fixation duration $\geq 500ms$ else NaN
Saccade	Saccadic amplitude	Euclidean distance to the previous fixation in the (code) line and column coordinates
	No saccade	<i>Expression:</i> 1 if source (i.e., previous) fixation is on the same line and column else 0
	Horizontal	<i>Expression:</i> 1 if source fixation is on the same line and different column else 0
	Non-horizontal	<i>Expression:</i> 1 if source fixation is on a different line else 0
Pupil	Pupil size <i>{aggr.}</i>	Pupil size in mm collected over the specified time window and aggregated using <i>{count, sum, max, min, mean, median, std}</i>
	Number of pupil peaks	Number of pupil peaks observed in the specified time window
	Amplitude of pupil peaks <i>{aggr.}</i>	Amplitude of the individual pupil peaks observed in the specified time window and aggregated using <i>{count, sum, max, min, mean, median, std}</i>
(sub) Scan-path	Sub-scan length – # transitions	Number of transitions between lines of code in the sub-scan derived over the specified time window. In our graph-based representation, this corresponds to the number of edges multiplied by their weights
	Sub-scan length – # unique visits	Number of unique lines of code visited in the sub-scan derived over the specified time window. In our graph-based representation, this corresponds to the number of nodes in the graph
	Sub-scan density	(# transitions / # unique line visits)
	Sub-scan entropy	Entropy of the sub-scan transition matrix. In our graph-based representation, this corresponds to the entropy of the graph adjacency matrix
Cluster-based AOIs	Is fix. in spatial cluster	<i>Expression:</i> 1 if fixation is in spatial cluster else 0
	Is fix. in spatial-temporal cluster	<i>Expression:</i> 1 if fixation is in spatial-temporal cluster else 0

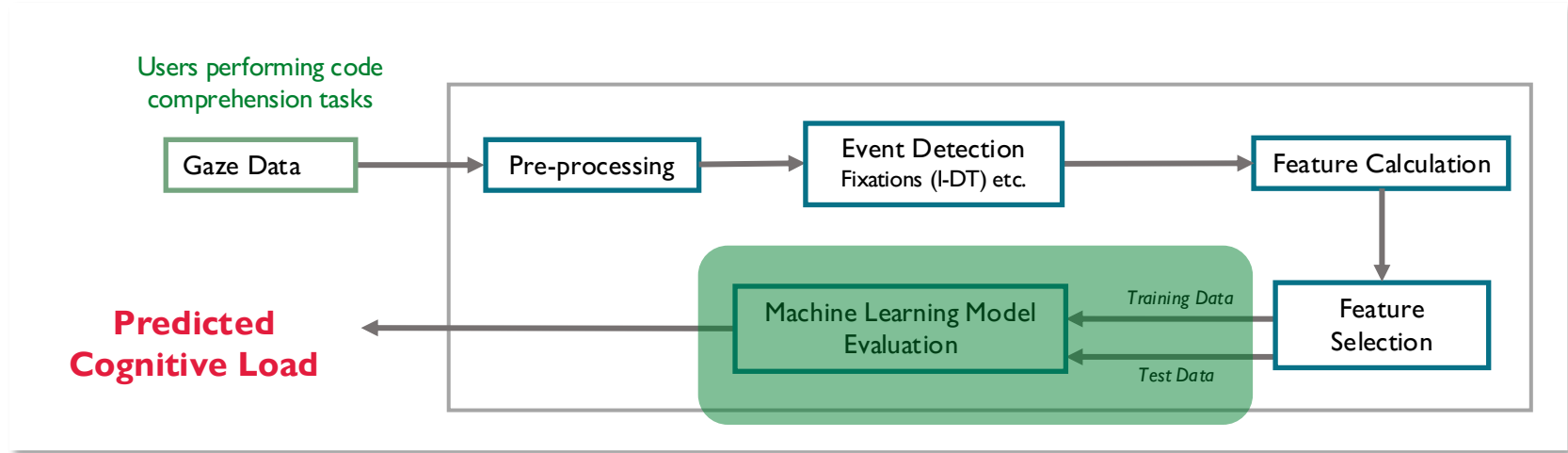
Features extracted at the **fixation level**. Abbreviations: Fix.: Fixation

Feature Calculation and Selection

Category	Feature	Description
Fixation	Fixation count	Number of fixations on the line of code
	Fixation count <i>fix.</i> $\geq 250ms$	Number of fixations with duration $\geq 250ms$ on the line of code
	Fixation count <i>fix.</i> $\geq 500ms$	Number of fixations with duration $\geq 500ms$ on the line of code
Cluster-based	# Spatial clusters	Number of spatial clusters formed on the line of code
AOIs	# Spatial-temporal clusters	Number of spatial-temporal clusters formed on the line of code

Additional features extracted at the **line level**. Abbreviations: Fix.: Fixation

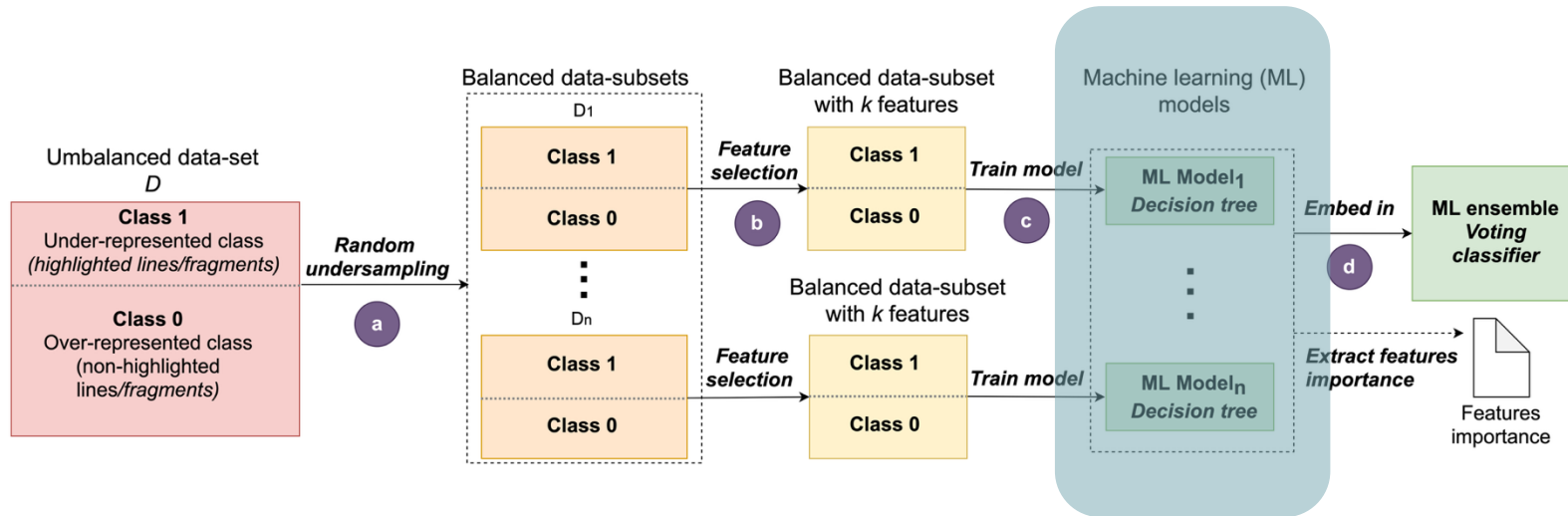
Back to Our Pipeline



Machine Learning Model Training

Configuration

- **Supervised:** Ensemble learning
- **Features:** Cognitive load measures collected at a fine-grained level
- **Labels:** Participants' highlights of the mentally demanding parts of code



Architecture of the ensemble learning model used

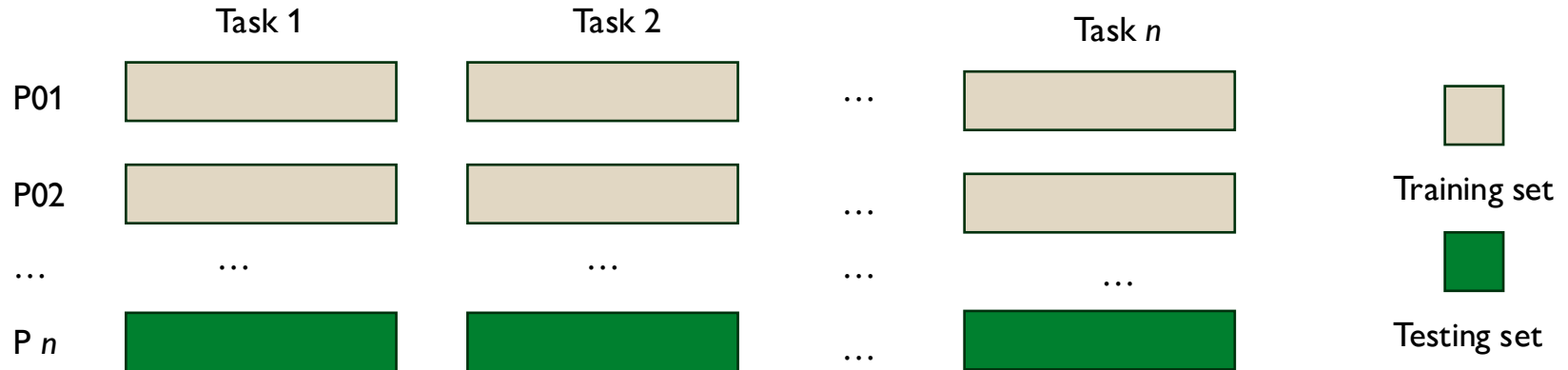
Machine Learning Model Evaluation

Cross-validation Variants

Leave one participant out

Leave one task out

Leave one participant and task out



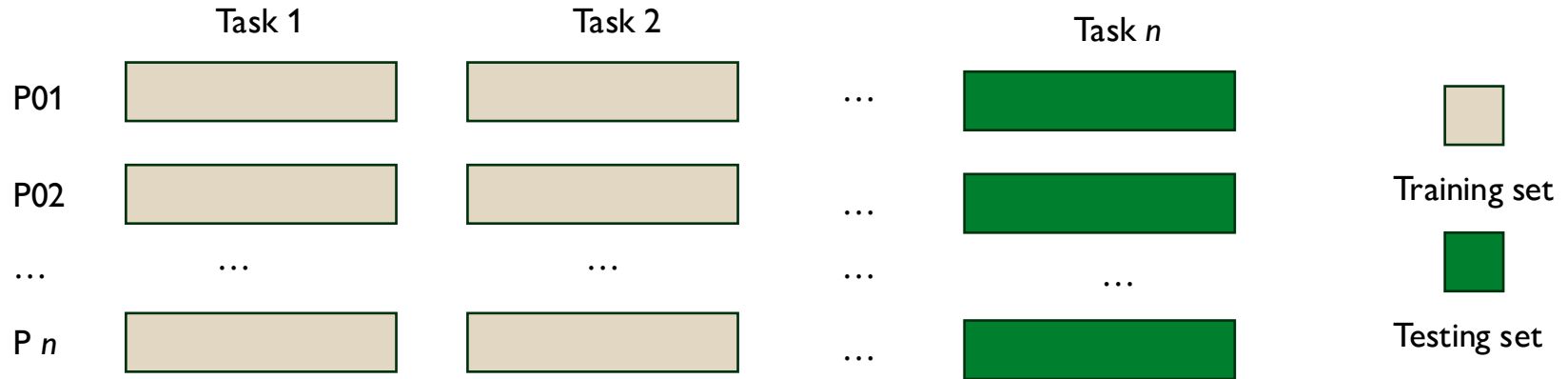
Machine Learning Model Evaluation

Cross-validation Variants

Leave one participant out

Leave one task out

Leave one participant and task out



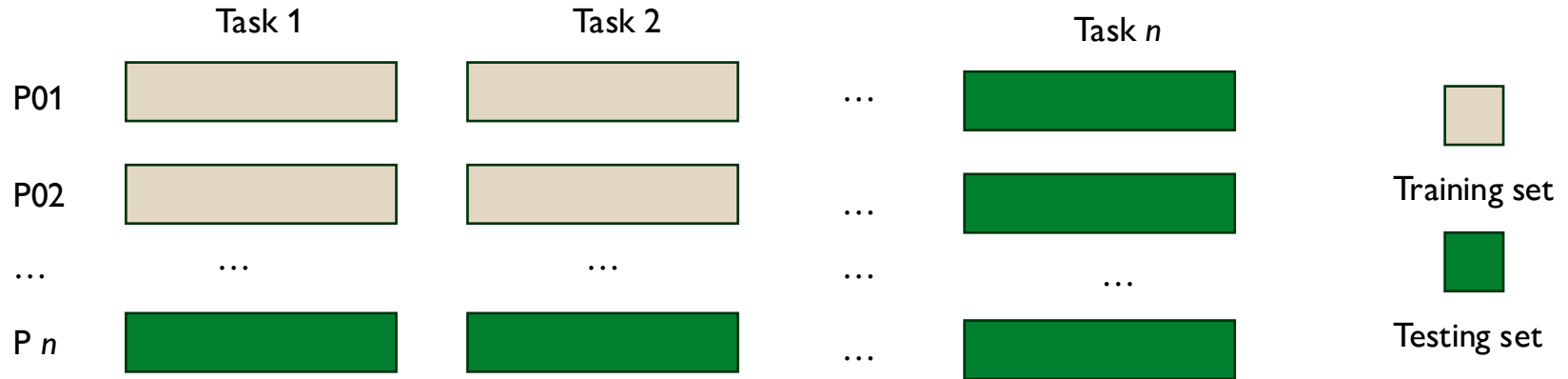
Machine Learning Model Evaluation

Cross-validation Variants

Leave one participant out

Leave one task out

Leave one participant and task out



Machine Learning Model Evaluation

Common Metrics

- **Accuracy:** Proportion of all predictions that are correct. $(TP + TN) / Total$
- **Precision:** Of all predicted positives, how many are actually positive. $TP / (TP + FP)$
- **Recall:** Of all actual positives, how many were correctly identified. $TP / (TP + FN)$
- **F1 Score:** Harmonic mean of Precision and Recall; balances both into a single metric. $2 \times (Precision \times Recall) / (Precision + Recall)$

Example of Prediction results following the cross-validation approach

Level	Highlights	One-participant-out				One-task-out				One-participant-task-out			
		F1	Recall	Acc.	Prec.	F1	Recall	Acc.	Prec.	F1	Recall	Acc.	Prec.
Line	Initial	67.51%	67.36%	69.94%	71.23%	65.17%	65.98%	67.92%	68.94%	64.74%	70.73%	70.80%	69.51%
Fragment	Initial	84.98%	85.47%	85.16%	86.25%	80.79%	80.86%	84.21%	85.29%	79.78%	82.84%	84.16%	81.15%
Line	Revised	68.57%	69.01%	70.75%	71.78%	67.22%	67.43%	70.30%	71.60%	65.96%	73.35%	71.48%	68.96%
Fragment	Revised	85.65%	84.25%	86.24%	88.61%	81.46%	80.26%	84.47%	85.82%	81.14%	84.32%	85.56%	82.21%

Machine Learning Model Evaluation

Feature Importance: A score indicating how much each input feature contributes to a model's predictions.

Higher importance means the feature has a stronger influence on the output.

Example of feature importance results

Feature gr.	Ac.	Level	Highl.	Feature gr. importance		
				OPO	OTO	OPTO
Fixation	LI	Line	Initial	0.102	0.126	0.132
	FI	Frag.	Initial	0.118	0.093	0.11
	LR	Line	Revised	0.155	0.162	0.167
	FR	Frag.	Revised	0.082	0.094	0.092
Saccade	LI	Line	Initial	0.134	0.151	0.157
	FI	Frag.	Initial	0.653	0.68	0.655
	LR	Line	Revised	0.126	0.135	0.115
	FR	Frag.	Revised	0.702	0.7	0.694
Pupil	LI	Line	Initial	0.587	0.524	0.547
	FI	Frag.	Initial	0.147	0.159	0.164
	LR	Line	Revised	0.539	0.563	0.553
	FR	Frag.	Revised	0.151	0.141	0.147
Scan-path	LI	Line	Initial	0.174	0.198	0.158
	FI	Frag.	Initial	0.076	0.062	0.064
	LR	Line	Revised	0.176	0.135	0.158
	FR	Frag.	Revised	0.059	0.055	0.062
Cluster-based AOI	LI	Line	Initial	0.003	0.002	0.006
	FI	Frag.	Initial	0.006	0.007	0.007
	LR	Line	Revised	0.004	0.004	0.007
	FR	Frag.	Revised	0.005	0.01	0.006

Offline Gaze Data Analysis for Activity Recognition

We provide you with

Gaze dataset (*for the activity detection usecase*)

- 20 participants
- 3 activities (Read, Inspect, Search)

Jupyter Notebooks

- Process and visualize gaze data
- Gaze Feature calculation
- SVM-based activity classifier

You can

Explore these notebooks

**And replace SVM
with another model**

Activity Prediction from Eye Movements

1. Data Collection

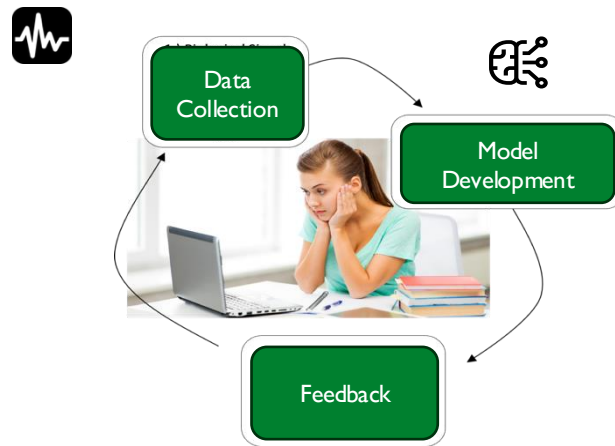
- Understand the activity: For example, *Reading* material (paper or digital etc.), geometric constraints, static vs. dynamic scene (scrolling) etc.
- Understand the temporal aspects that are relevant for data collection and model development (seconds to hours)
- Collect ground truth data

2. Model Development

- Calculate eye movement metrics (fixation, saccade, blinks, etc.)
- Data Preprocessing (cleaning, smoothing, interpolation, etc.)
- Calculate features (normalization)
- Split data to training and test sets
- Train models and document their performance

3. Feedback

- What do we do with the detected activity?



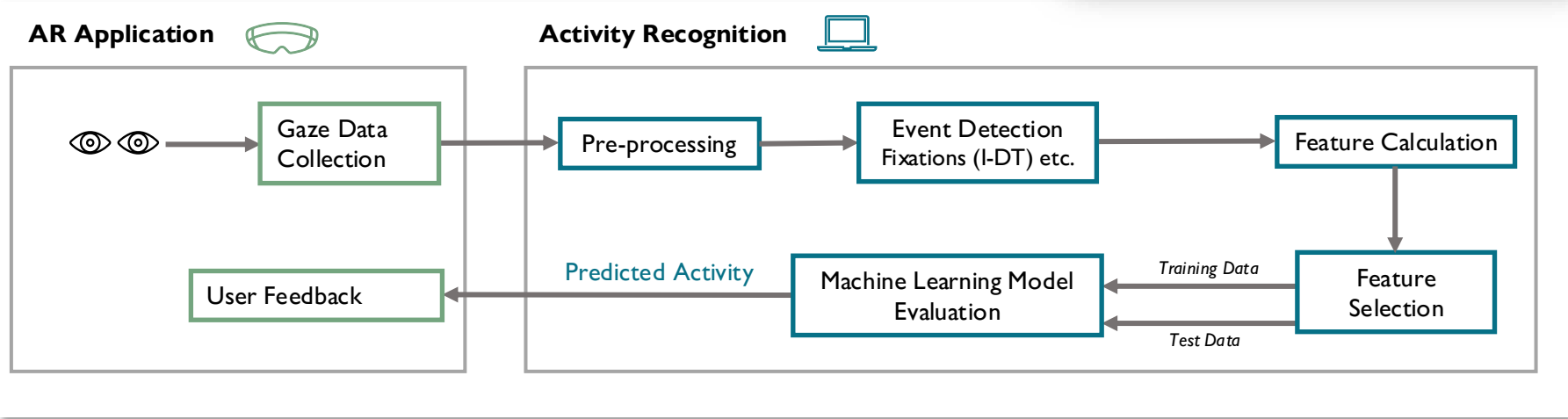
Augmented Reality Feedback



You will simulate a working online LLM-based feedback to the user that runs on the notebook.

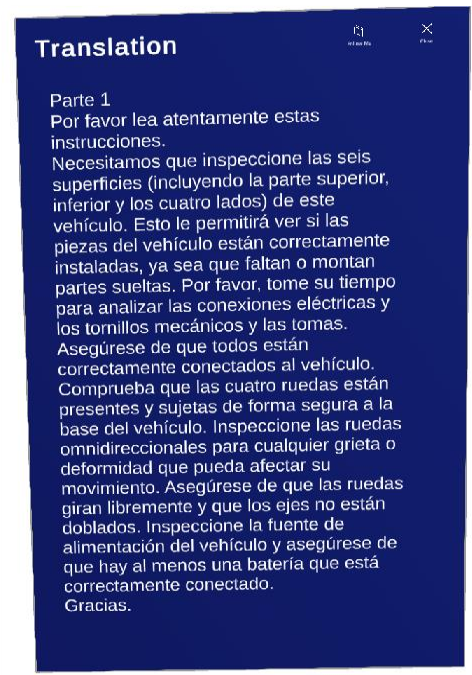
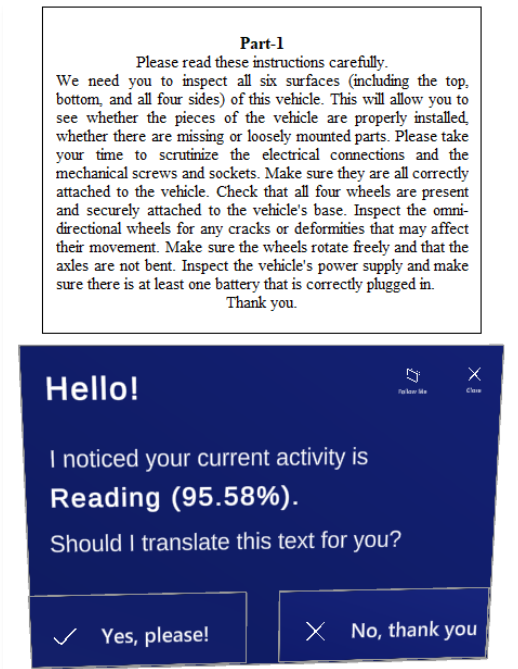
The feedback must be relevant to the current activity and provide some service linked with that activity.

Pipeline:



Augmented Reality Feedback (example)

For the **Reading** activity, e.g.,
translating the text.



Notebook 4:



📁 Data/RawGazeData	Delete Data/RawGazeData/placeholder.txt
📁 demo_images	intervention example added
📄 AnSVMClassifierForHL2GazeFeatures.ipynb	Add files via upload
📄 FeatureCalculation.ipynb	Add files via upload
📄 InterventionExample.ipynb	intervention example added
📄 OnlineFeaturesCalculation.ipynb	Add files via upload
📄 README.md	Initial commit
📄 ReadAndVisualizeRawHL2GazeData.ipynb	Add files via upload

Experimenting with Different LLMs

```
response = client.chat.completions.create(  
    model="openai/gpt-4o",  
    messages=  
        {  
            "role": "user",  
            "content": [  
                {  
                    "type": "image_url", "image_url": {"url": f"data:image/png;  
                    base64,{b64_image}"},  
                },  
                {  
                    "type": "text", "text": PROMPTS[detected_task]},  
            ],  
        },  
    ],  
)
```

InterventionExample.ipynb

You need to choose a multi-modal LLM that accepts both text and images

Check for available models in OpenRouter and use the API key that we will provide to you

OpenRouter

Search

Home Models Fusion Chat Rankings Apps Docs Personal

Input Modalities

Text Image File Audio Video

Context length

Prompt pricing

Series

Categories

Supported Parameters

Distillable

Zero Data Retention

In-Region Routing

Providers

Models

openai

Compare

Top Weekly

OpenAI: GPT-5.5

490B tokens

Academia (#30) Finance (#31) Health (#26) SEO (#41) Programming (#11) +3 categories

GPT-5.5 is OpenAI's frontier model designed for complex professional workloads, building on GPT-5.4 with stronger reasoning, higher reliability, and improved token efficiency on hard tasks. It features a 1M+ token context window (922K input, 128K output) with support for text and image inputs, enabling large-scale reasoning, coding, and multimodal workflows within a single system.

by gpt4all Apr 24, 2025 1.05M context \$5.00/M input tokens \$30.00/M output tokens

OpenAI: gpt-oss-120b

441B tokens

Academia (#9) Finance (#15) Health (#12) Legal (#3) Marketing (#2) +7 categories

gpt-oss-120b is an open-weight, 117B-parameter Mixture-of-Experts (MoE) language model from OpenAI designed for high-reasoning, agentic, and general-purpose production use cases. It activates 5.1B parameters per forward pass and is optimized to run on a single H100 GPU with native MXFP4 quantization. The model supports configurable reasoning depth, full chain-of-thought ...

by gpt4all Aug 5, 2025 131K context \$0.039/M input tokens \$0.18/M output tokens

OpenAI: GPT-5.4

273B tokens

Academia (#16) Finance (#23) Health (#18) Legal (#29) Marketing (#14) +5 categories

GPT-5.4 is OpenAI's latest frontier model, unifying the Codex and GPT lines into a single system. It features a 1M+ token context window (922K input, 128K output) with support for text and image inputs, enabling high-context reasoning, coding, and multimodal analysis within the same workflow. The model delivers improved performance in coding, document understanding, tool use, and ...

by gpt4all Mar 5, 2025 1.05M context \$2.50/M input tokens \$15.00/M output tokens

OpenAI: GPT-4o-mini

266B tokens

Academia (#12) Finance (#6) Health (#11) Legal (#15) Marketing (#11) +7 categories

Pay attention to the allowed modalities and cost

LLM-based User Feedback



We provide you with

A Notebook that can take

An activity

An image of the current scene

An example prompt

To generate a relevant feedback

You can

Explore this notebook

**And replace the input image and
existing prompt to explore the
potential of the current pipeline**

Interactive Session: Task Overview

Goal

Design an adaptive system that detects a user state or activity and delivers a meaningful intervention.

By the end of the session, each group should have:

- A clearly defined adaptive system for a particular user group and use case
- A context prediction (e.g., activity, cognitive load, affective state) approach
- An eye-tracking data collection and evaluation strategy
- An intervention/ feedback approach
- A 3 to 5 minutes presentation summarizing their entire system (e.g, slides, storyboards, flowcharts focusing on the system workflow)

Interactive Session: Value Framing (10 mins)

In groups

- Think of a **domain** where developing an adaptive system is relevant (e.g., e-learning, software development, healthcare or any domain ***possibly related to your research or job activities***)
- Define the **value** of your adaptive system.
 - **Users:** Who uses or is affected by the system?
 - **Purpose:** What problem does the system address?
 - **Context of use:** When and where is the system used?
 - **Adaptation need:** Why is adaptation better than a fixed interface or fixed feedback?
 - **Value delivered:** What should improve for the user, task, or organization?

Interactive Session: Sensing and Prediction (10 mins)

In groups

- Define **what the system detects** when adaptation is needed
 - **Target state or activity:** What do you want to detect?
Examples: confusion, cognitive load, loss of attention, wrong strategy, task progress.
 - **Input data:** What measures are needed? Gazes + what?
Examples: clicks, scrolls, task events, text input, time on task, emotions, face gestures
 - **Labels:** What ground truth would you need for training and evaluation? and how would you obtain them?
 - **Collection settings:** Where would the data be collected?
Examples: Lab, classroom, workplace, online system, simulator
 - **Model evaluation:** How would you know that the detection model works well?

Interactive Session: Intervention Design (10 mins)

In groups

- Define **what the system does after detecting a relevant state or activity**.
 - **Trigger:** What detection result activates the intervention?
 - **Timing:** When should the intervention appear?
 - **Intervention type:** What does the system provide?
Examples: hint, warning, explanation, simplification, visualization, task recommendation, sound, vibrotactile
 - **Personalization:** How would the intervention adapt to the user or situation?
 - **Risk Control:** When should the system avoid intervening?

Interactive Session: LLM-based Intervention (15 mins)

In groups

- Explore **how an LLM could generate the intervention**.
- Create and test 2–3 prompts using the provided notebook.
- For each prompt, specify:
 - User context
 - Detected state or activity
 - Available data (e.g., screen capture, environment image)
 - Desired intervention style
 - Constraints on the response
- Then reflect briefly:
 - Was the response useful?
 - Was it too generic?
 - Did it respect the user context?
 - What information was missing?
 - What risks or failure cases appeared?

Interactive Session: System Flow and Final Presentation (15 mins)

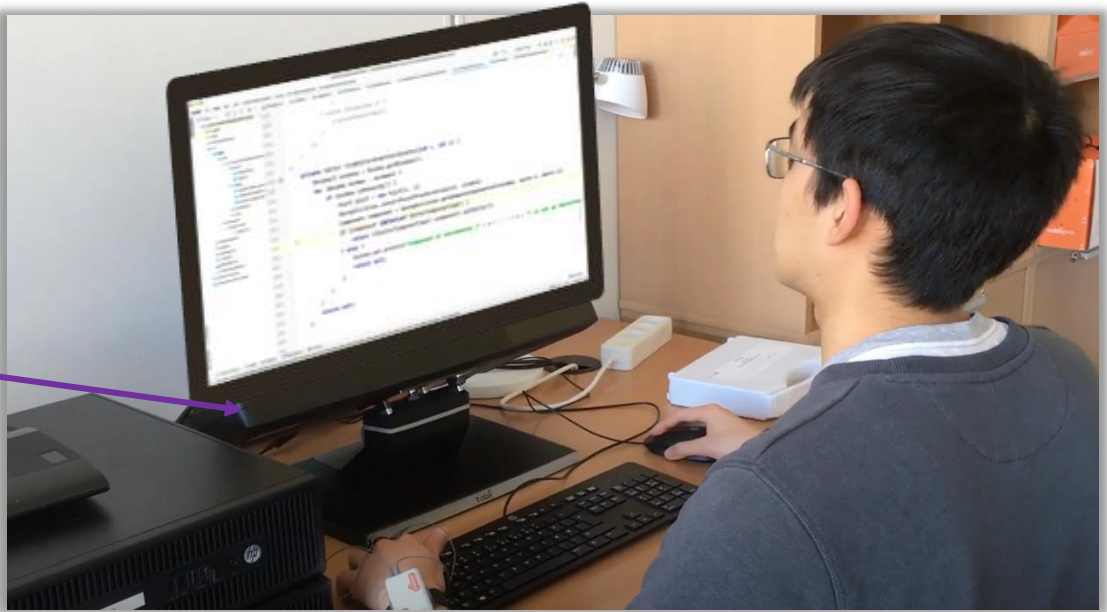
In groups

- Draw a **flowchart or storyboard** of your adaptive system.
- Your flow should include:
 - User performs a task
 - System collects interaction or sensor data
 - Features are extracted
 - Model predicts user state or activity
 - System decides whether to intervene
 - Intervention is delivered
 - User reacts
 - System logs the outcome for evaluation or improvement
- Add your flow to a **presentation** covering the following
 - The domain and target users
 - The problem addressed
 - The data and prediction model
 - The intervention mechanism
 - How the system flow works
 - How you would evaluate success

Example Idea:

*While working on a large code base (e.g., Java project),
automatically detect the parts of the source code that are difficult to understand and
provide the user contextually relevant feedback*

A screen-based
Eye-tracker



Example Idea

The screenshot shows an IDE window titled "contextAdaptiveSystem - HttpMethodSelector.java [contextAdaptiveSystem.main]". The code is a Java file with the following structure:

```
231 }
232 } catch (Exception e) {
233     e.printStackTrace();
234 }
235 });
236 }
237 */
238 @ private Editor findEditorUnderCoordinates(int x, int y) {
239     Window[] windows = Window.getWindows();
240     for (Window window : windows) {
241         if (window.isShowing()) {
242             Point point = new Point(x, y);
243             SwingUtilities.convertPointFromScreen(point, window);
244             Component component = SwingUtilities.getDeepestComponentAt(
245                 window, point.x, point.y);
246             if (component instanceof EditorComponentImpl) {
247                 return ((EditorComponentImpl) component).getEditor();
248             } else {
249                 System.out.println("Component at coordinates (" + x + ", " + y + ") is not an EditorComponentImpl");
250                 return null;
251             }
252         }
253     }
254     return null;
255 }
```

A red dashed box highlights a section of the code, specifically the `findEditorUnderCoordinates` method. To the right of the code, a red text box contains the following text:

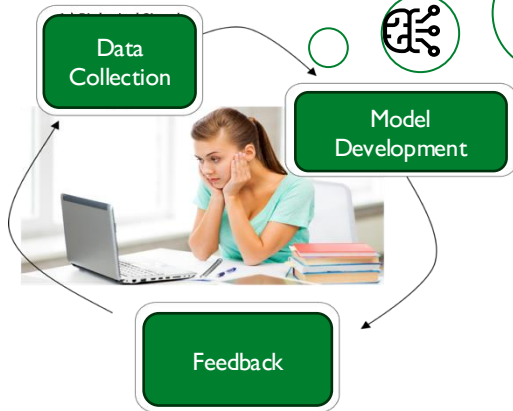
High cognitive load was detected when reading `findEditorUnderCoordinates()`!

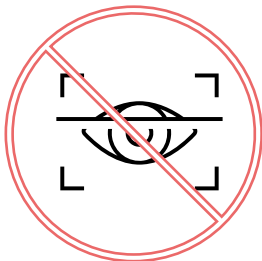
The following is a brief explanation of this method:

- **Retrieve Windows:** The method gathers all windows currently associated with the application.
- **Filter Visible Windows:** It then checks each window to see if it is currently visible on the screen.
- **Coordinate Conversion:** The screen coordinates provided are converted into coordinates relative to each window being checked.
- **Find Deepest Component:** Within each visible window, the method finds the most deeply nested component at the specified coordinates.
- **Check Component Type:** It checks if this component is of a specific type (`EditorComponentImpl`).
- **Return Editor:** If the component is the expected type, the method retrieves and returns the associated `Editor` object.
- **Handle Non-Matching Component:** If the component is not the expected type, it logs a message and returns `null`.

At the bottom of the red dashed box, there are two buttons: "Add comments to" and "Refactor the code".

Do **You** use Glasses or Contact Lens?

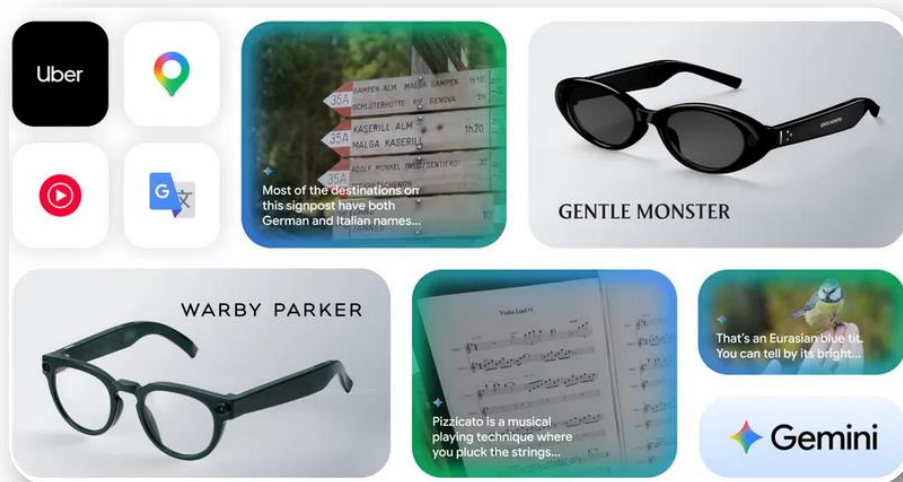




Meta Ray-Ban Display



Google's Gentle Monster and Warby Parker



Thank You!

Dr. Kenan Bektaş

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programming-and-development/](https://ics.unisg.ch/chairs/barbara-weber-software-systems-programming-and-development/)

